

B.Sc. (Hons.) Semester IV Examination, 2013-14
GEOLOGY

Paper: GLB-401: Palaeontology and Stratigraphy

Time: Three Hours

Full Marks: 70

Note: Answer five questions in all, selecting two each from Section-A and Section-B. Section-C is compulsory. All questions carry equal marks.

SECTION-A

1. Define Palaeontology and describe its historical and evolutionary relationship with other operational branches of geology.
 2. Briefly discuss any **two** of the following concepts:
 - (a) The essential biological and physical conditions of fossilization
 - (b) The theories on the scientific origin of life
 - (c) Various ecological terms used to describe the bathymetric distribution of marine organisms
 3. Describe the comprehensive morphological features of Trilobites. Support your descriptive answer with neat, systematically labeled structural diagrams.
 4. Write descriptive short notes on any **two** of the following:
 - (a) Incompleteness of the fossil record and taphonomic bias
 - (b) The architecture of the Geological Time Scale
 - (c) Economic importance and rock types of the Cuddapah Group
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SECTION-B

1. Describe in detail the various fundamental principles of stratigraphy. Illustrate your answer using neat, conceptual cross-sectional diagrams.
 2. Describe in detail the geographic distribution, lithostratigraphic classification, and economic importance of the Dharwar Supergroup. Briefly discuss the specific types of geological information derived from studying these ancient rocks.
 3. Write comprehensive short notes on any **two** of the following:
 - (a) In what way stratigraphy fundamentally requires multi-disciplinary inputs from all branches of geology
 - (b) What is the evolutionary and structural significance of the Eparchaeon Unconformity
 - (c) Why structural unconformities in the field are frequently marked by basal conglomerates
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SECTION-C

Compulsory Section: Distinguish between the following pairs or answer the prompts in 1 to 3 sentences each:

1. Differentiate between *fossils* and *pseudofossils*.
 2. Differentiate between a *species* and a *subspecies*.
 3. Define the stratigraphical limits of the *Sargur Group*.
 4. Differentiate between *chondrites* and *dendrites* in sedimentary structures.
 5. Define the geological significance of a *diamondiferous conglomerate*.
 6. What are the formal definitions of *lithostratigraphic units*?
 7. Define the position and role of the *apical disc* in Echinoids.
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ExternalPortals&AcademicResources

- **ExaminationPortal:** <https://bhu-exam.vercel.app/>
- **StudentResourceCenter:** <https://quashan-me.vercel.app/>
- **AcademicPortfolio:** <https://me-aryan.vercel.app/>